Clear-Air Anomaly Masking Using Kalman Temporal Filter From Geostationary Multispectral Imagery

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***Abstract*— An anomaly detection and estimation technique, using a pixel-based modified version of the Kalman temporal filter, is presented. To detect the anomalies of the observed variable, the proposed Kalman-based anomaly masking (KAM) algorithm relies on the background state models of the expected measurement cycle of each pixel in nominal (abnormal) conditions. If the measurement significantly deviates from its expected value as predicted by *a priori* state, an anomaly is identified. The KAM algorithm also provides an *a priori* estimate of the nominal scenario, exploiting the previous Kalman filter states. The product is an equivalent clear-air observation, expected to be measured in the absence of anomaly (e.g., in the absence of cloud coverage). The KAM algorithm exhibits a general applicability, since its estimates are empirically computed from pixel-based models and its thresholds can be set independently of the area of interest. An application of the KAM algorithm to clear-air nominal scenarios is shown using multispectral imagery from the geostationary Spinning Enhanced Visible and Infrared Imager, having 12 visible-infrared channels and repeat cycle of 15 min, on board of the Meteosat satellite. The area of interest covers the whole West Africa for a test period of three months (December 2015 until February 2016). This results in a massive amount of processed pixels (i.e., 1530 × 880 pixels for 96 timeslots per day). A validation of the clear-air KAM algorithm is presented by intercomparing the detection results with the well-known EUMETSAT cloud mask product. The validation shows constant percentages of matching around 90% over the entire period of analysis.**

***Index Terms*— Anomaly detection, clear-air masking, environmental monitoring, geostationary satellite observations, global-scale analysis, Kalman filtering, visible-infrared (VIS-IR) measurements.**

I. Introduction

HAZARD and meteorological applications continuously rely on remote-sensing techniques and, in the past decades, the international scientific research community has recognized the satellites to be vital in detecting and monitoring natural and environmental phenomena. In several countries or critical situations of emergency, satellite observations are the only economically feasible way to detect and track these phenomena, and eventually mitigate the repercussions [1].

Space-borne radiometers offer the possibility to sense the environment by exploiting the thermal infrared (IR) and optical channels [4], [5]. These measurements are obtained from both geosynchronous-Earth-orbit (GEO) and low-Earth-orbit (LEO) satellites, thus offering different spatial and temporal resolutions [4]–[17]. This translates into having different and complementary advantages, and the best instrument to be used strongly depends on the considered application [12].

For nearly real-time applications, geostationary platforms often are a mandatory choice as they are able to observe the same Earth scene with a rapid and continuous sampling rate [2]. However, given the large distance from Earth, they have the disadvantage of a low spatial resolution (typically larger than a few kilometers). On the other hand, for LEO payloads, the revisit time may be even longer than 12 h but they claim very high spatial resolutions that vary from several kilometers down to meters, depending on the wavelength used by the sensor aboard the platforms [4].

Although the temporal availability of GEO imagery is quite high, many operational services require small-time resolutions with relatively small data latency to operate in near real time. Many works have stated the importance of utilizing operational meteorological satellites for the retrieval of routine products and long-term analyses [1], showing the possibility of developing detection algorithms using GEO satellites [1]–[30]. Most of the mentioned approaches commonly used are designed on spatially-based algorithms, that is they consider a set of spatial thresholds to be contextually applied in absolute or differential way with respect to the averaged scene around the pixel of interest [4], [5]–[23]. Following a purely spatial contextual algorithm could be disadvantageous in certain conditions, as the temporal trend information is not exploited to add robustness to the system (especially when a time correlation exists between two consecutive observations) [31]–[35].

This article aims at defining a generalized remote-sensing technique, which implements a modified version of the well-known Kalman filter [8]–[11], [13]. Roughly speaking, the Kalman filter is a recursive algorithm able to estimate the true state of a system on the basis of noisy measurements

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and has long been considered the optimal solution to many tracking and data prediction tasks [13], providing linear, unbiased estimations of the states of a system. Historically, it has been used in a large variety of applications [11]–[13] and it is still a very popular solution in several science fields, such as supervisory management systems [6], as well as data fusion [7], as ones of the most recent examples. The Kalman filter typically works in a time-based domain and is able to take into account multiple representations of the same observed scene (e.g., multispectral measurements). In this article, we provide a detailed formulation of a Kalman-based anomaly masking (KAM) technique, which can suit most of the Earth observation and remote-sensing applications characterized by recurrent observations of the same scene in a reasonably short time interval [14], [15]. The possibility of exploiting the time-evolution information of the observations is indeed one of the elements of novelty of the presented approach, adding robustness to the detection algorithm when persistent anomaly conditions are observed for a relatively large area in space (critical condition for spatial-based detection algorithms (see [5]–[23]).

As real example of application, the KAM approach is applied to detect clear-air anomalies, making use of GEO visible-infrared (VIS-IR) passive measurements from the Meteosat Second Generation (MSG) satellite. We follow a purely temporally-based approach to detect abnormal behaviors of the spectral radiance [14]. The clear-air KAM estimate relies on a background model of the daily measurement cycle of the considered pixel in pure clear-sky conditions. If the measurement significantly deviates from its expected value as predicted by the algorithm (*a priori* state), the system identifies the presence of an anomaly, which is classified as a non-clear-air scenario. The difference between the actual scenario and the corresponding *a priori* Kalman estimates is therefore the main driver. Practically speaking, a clear-air classification only happens if the measurements strictly satisfy several multispectral conditions, which refer to the *a priori* estimates and the pixel models. The filter updates the *a posteriori* states depending on the detection outcome: the clear-sky model dominates only in the presence of an anomaly; otherwise, the filter follows the actual measurements.

An important add-on value of the KAM approach is to be able to provide *a priori* estimates of the scenario, exploiting the previous Kalman filter states and making use of the pixel models. This allows having an equivalent clear-air radiance which is expected to be measured in the absence of anomaly (for instance, in the absence of cloud coverage). In such a way, the algorithm results globally applicable, since the estimates are computed from the pixel models and the thresholds can be set independently of the area of interest.

This article is structured as follows. Section II presents the mathematical formulation of the modified Kalman filtering technique, which can be easily tailored to accommodate several applications. Section III describes the GEO measurements used to test and validate the approach, whereas Section IV provides the detailed information on the actual filter tailoring for clear-air masking. Eventually, Section V shows an intercomparison in terms of clear-air detection with respect to a highly assessed Cloud-Mask product using the same input measurements. Section VI draws the conclusion plus several considerations on future works and possible application developments.

TABLE I

MODIFIED TEMPORAL KALMAN FILTER, SUMMARY OF VARIABLES, AND THEIR DEFINITIONS WITH n THE NUMBER OF AVAILABLE OBSERVATIONS SPANNED BY THE INDEXES $i = 1, \ldots, n$; $j = 1, \ldots, n$

Variable	Definition	Dimension
$\boldsymbol{z}_k$	True observation vector	$(n\times 1)$
$\hat{\boldsymbol{z}}_k$	Actual observation vector (known)	$(n\times 1)$
$\boldsymbol{\varepsilon}_k$	Associated observational error	$(n\times 1)$
$\boldsymbol{R}_k$	Observation noise covariance matrix (known)	$(n\times n)$
$\boldsymbol{m}_k$	Background model vector (known)	$(n\times 1)$
$\boldsymbol{\sigma}_k^m$	Background deviation vector (known)	$(n\times 1)$
$\boldsymbol{A}_k$	State transition matrix	$(n\times n)$
$\boldsymbol{Q}_k$	Process noise covariance matrix	$(n\times n)$
$\boldsymbol{x}_k^-$	Kalman a-priori state vector	$(n\times 1)$
$\boldsymbol{x}_{k-1}^+$	Kalman a-posteriori state vector	$(n\times 1)$
$\boldsymbol{P}_k^-$	Kalman a-priori error covariance matrix	$(n\times n)$
$\boldsymbol{P}_{k-1}^+$	Kalman a-posteriori error covariance matrix	$(n\times n)$
$\boldsymbol{K}_k$	Kalman gain matrix	$(n\times n)$
$\boldsymbol{t}_k$	Anomaly threshold dynamic vector	$(n\times 1)$
$\boldsymbol{t}_m$	Maximum threshold vector	$(n\times 1)$
$\boldsymbol{\sigma}^t$	Anomaly confidence interval vector	$(n\times 1)$

II. MODIFIED TEMPORAL KALMAN FILTER

Hereby we present a modified version of the well-known Kalman filter [13]. Such filter can theoretically suit most of the Earth observation and remote-sensing applications characterized by recurrent observations of the same scene in a reasonably short time interval. The proposed KAM filter works in a time-based domain and is able to take into account multiple representations of the same observed scene (e.g., multispectral measurements).

In the following paragraphs, we assume time-discrete observations equally spaced by a constant time sampling T_s. Moreover, a daily cycle T_c is assumed. The index k indicates the considered time step $k = 1, \ldots, T_s/T_c$. The filter can be easily adapted to operational situations in which the time sampling is not uniform, as long as the measurements are recurrent, that is the correlation between two consecutive observations is not null. The Kalman nomenclature and the adopted vector notation are summarized in Table I.

A. Kalman Theoretical Background

We consider a number of n representations of the same scenario (e.g., a multispectral characterization of the scene

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with n observations). The actual observation vector $\hat{\boldsymbol{z}}_k$ ($n \times 1$) is composed by the observed values of the scene at the instant k (e.g., measured radiances at n different frequencies as measured by the sensors)

$$\hat{\boldsymbol{z}}_k = \boldsymbol{z}_k + \boldsymbol{\varepsilon}_k = \begin{pmatrix} z_1 + \varepsilon_1 \\ z_2 + \varepsilon_2 \\ z_3 + \varepsilon_3 \\ \vdots \\ z_n + \varepsilon_n \end{pmatrix}_k = \begin{pmatrix} \hat{z}_1 \\ \hat{z}_2 \\ \hat{z}_3 \\ \vdots \\ \hat{z}_n \end{pmatrix}_k \tag{1}$$

where $\boldsymbol{z}_k$ ($n \times 1$) is the true observation vector, containing the true values of the scene, whereas $\boldsymbol{\varepsilon}_k$ represents the observational error associated with each measurement. By definition, the measurements $\hat{\boldsymbol{z}}_k$ are affected by various error sources (random and/or systematic) and they could deviate from their true value $\boldsymbol{z}_k$ where the errors being possibly cross-correlated [21]. Hence, the need of defining the covariances among noise pairs at the timeslot k, forming the observation noise covariance matrix $\boldsymbol{R}_k$ ($n \times n$) defined as follows:

$$\boldsymbol{R}_k = \begin{pmatrix} R_{11} & R_{12} & R_{13} & \cdots & R_{1n} \\ R_{21} & R_{22} & R_{23} & \cdots & R_{2n} \\ R_{31} & R_{32} & R_{33} & \cdots & R_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ R_{n1} & R_{n2} & R_{n3} & \cdots & R_{nn} \end{pmatrix}_k . \tag{2}$$

The covariances among noise pairs, such as elements $\{R_{ij}\}_k$, can be computed as follows:

$$\{R_{ij}\}_k = \{E\left[\left(\hat{z}_i - z_i\right)\left(\hat{z}_j - z_j\right)\right]\}_k \tag{3}$$

where $\hat{z}_{i,j}$ and $z_{i,j}$ are elements of the actual and true observation vectors, respectively. The operator E is the expected value operator and the indexes $i = 1, 2, \ldots, n$ and $j = 1, 2, \ldots, n$ refer to rows and columns of the matrix, respectively. In other words, the observation noise covariance matrix $\boldsymbol{R}_k$ defines possible mutual correlations among mean observational errors, associated with the measurements.

The state transition matrix $\boldsymbol{A}_k$ ($n \times n$), also known as fundamental matrix, defines how two consecutive states are linked to each other [13]. There are several ways to define and retrieve the best fundamental matrix for a given application, often making use of Laplace transforms or Taylor series expansions. Once we have the fundamental matrix $\boldsymbol{A}_k$ that fits our application, we can simply propagate states forward by matrix multiplication, by properly taking into account the sampling time [8]. For polynomial Kalman filters, the state transition matrix is often a function of time, computed by applying a polynomial expansion or a best-fitting technique over a consistent set of data [9]. For our purposes, dealing with discrete processes, we can compute the state transition matrix in a variety of ways, depending on the number of considered channels as well as their correlation, that is whether considering cross-channel relationships would increase robustness or not. In general, the state transition matrix can be obtained by exploiting *a priori* models already available or exploiting pixel observation models characterizing the nominal behavior of a certain observed scenario. Of course, a mix of the two approaches can also be considered for some applications, such as physically-based models, which need a set of nominal observations to fine-tune their trend.

In many applications, the fully empirical approach is the only possible one, especially for global-scale applications in which finding a global applicable physical model results in a challenging and hazardous choice. We can therefore exploit a background model vector of the considered scenario $\boldsymbol{m}_k$ ($n \times 1$), describing what one would expect if the observed scenario were nominal. The background model is to be intended as an independent source of information, which can be generally applicable, as a global-scale generic model, or even more accurate, for example, seasonal pixel models for given geographical areas. A nominal scenario is identified as, for instance, a set of measurements converging to a well-defined mean value (e.g., radiances in clear-air conditions) with as less associated noise as possible. In an ideal world, the background model vector corresponds to both actual and true observation vectors if the scenario is nominal.

For the scope of this article, let us suppose a linear relationship between two consecutive states and only consider the relationship only between the representations i of two consecutive states k and $(k-1)$, i.e., $\boldsymbol{A}_k$ is a diagonal matrix. In so doing, the elements of the state transition matrix depend on the background model vectors as follows:

$$\begin{cases} \{A_{ii}\}_k = \dfrac{\{m_i\}_k}{\{m_i\}_{k-1}} \\ \{A_{ij}\}_k = 0 \end{cases} \tag{4}$$

with $i = 1 \div n$, $j = 1 \div n$ and $i \neq j$. The state transition matrix is therefore strictly dependent on the considered timeslot, as it is directly related to the ratio between the background model m_i at a given discrete time k and its previous value at $(k-1)$. This approach is considered robust when the time resolution of the background model is dense enough to get an advantage in supposing a linear relationship among consecutive values. If this is not the case, the state transition matrix shall be expressed as a function of time, by adopting a polynomial relationship which best represents the model of interest.

The process noise covariance matrix $\boldsymbol{Q}_k$ ($n \times n$) is directly linked to the state transition matrix and it is a measure of the associated noise to each state transition. If a physical model is used, the matrix should be retrieved from its mathematical formulation. If the fully empirical approach in (4) is adopted, $\boldsymbol{Q}_k$ can be retrieved making use of the noise associated with the background model vector by applying the error propagation theory. If the aforementioned assumptions [and therefore (4)] hold, $\boldsymbol{Q}_k$ is also diagonal with $\{Q_{ij}\}_k = 0$ and its diagonal elements correspond to the estimated system variance, dependent on both background model and its standard deviation as follows:

$$\begin{cases} \{Q_{ii}\}_k = \left|\{A_{ii}\}_k\right|^2 \left[\left(\dfrac{\{\sigma_i^m\}_k}{\left|\{m_i\}_k\right|}\right)^2 + \left(\dfrac{\{\sigma_i^m\}_{k-1}}{\left|\{m_i\}_{k-1}\right|}\right)^2\right] \\ \{Q_{ij}\}_k = 0 \end{cases} \tag{5}$$

with $i = 1 \neq n$, $j = 1 \neq n$, and $i \neq j$, whereas the elements of $\boldsymbol{\sigma}_k^m$ are the standard deviation of the corresponding elements of $\boldsymbol{m}_k$. The equation above corresponds to the computation of

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the variance associated with $\{A_{ii}\}_k$, retrieved by adopting the uncertainty propagation theory applied to the ratio model [22].

B. Kalman A Priori States and Residuals

The Kalman *a priori* state vector approximates the observed scenario using *a priori* information available from the previous time slot. It describes the equivalent nominal expected scenario one would observe if no perturbations are present. The Kalman *a priori* state vector $\boldsymbol{x}_k^-$ $(n \times 1)$ at time k is defined as follows [9]:

$$\boldsymbol{x}_k^- = \boldsymbol{A}_k \boldsymbol{x}_{k-1}^+ \tag{6}$$

where $\boldsymbol{x}_{k-1}^+$ is the Kalman *a posteriori* state vector $(n \times 1)$ of the previous timeslot $k-1$. Similarly, the Kalman *a priori* error covariance matrix gives an estimate of the achieved accuracy for a given state at time step k by means of [9]

$$\boldsymbol{P}_k^- = \boldsymbol{A}_k \boldsymbol{P}_{k-1}^+ \boldsymbol{A}_k^{\mathrm{T}} + \boldsymbol{Q}_k \tag{7}$$

where $\boldsymbol{P}_{k-1}^+$ is the Kalman *a posteriori* covariance matrix of the previous time step $k-1$. The process noise covariance matrix $\boldsymbol{Q}_k$, representing the noise associated with the state transition matrix at a given timestamp, is added on top to account for the uncertainty introduced by the state propagation among consecutive discrete timeslots. It is to be noted that the *a priori* quantities described in this section are an important add-on information often not provided by classic spatial-based remote-sensing algorithms. Possible usages of these parameters are described in the conclusion.

Once computed the *a priori* states, we focus on the computation of the so-called residual quantities. These are simply a measure of how large is the discrepancy between the actual observed values and the *a priori* estimates. In many practical remote-sensing applications, having large residual quantities does not mean the algorithm is not working fine, contrary to the classic Kalman filtering theory.

Indeed, when observing a real physical phenomenon, we are actually interested in studying situations which go beyond the expected behavior and the presented algorithm aims at detecting and studying these phenomena. The residuals associated with the observation and covariance can be evaluated as follows:

$$\begin{aligned} &\text{Observation residual}: \hat{z}_k - \boldsymbol{x}_k^- \\ &\text{Covariance residual}: \boldsymbol{P}_k^- + \boldsymbol{R}_k. \end{aligned} \tag{8}$$

C. Anomaly Detection

We are here introducing the concept of anomaly intended as an observed scenario not complying with the nominal behavior of interest. In other words, our goal is to perceive if the actual observation vector can be classified as useful information for the filter propagation or it somehow deviates in such a way not to be beneficial to the following Kalman state. By only exploiting the *a priori* estimates and the background model, we consider a condition defining an anomaly-free region, which a nominal scenario must satisfy. If the scenario is a nonnominal state, then it is classified as an anomaly.

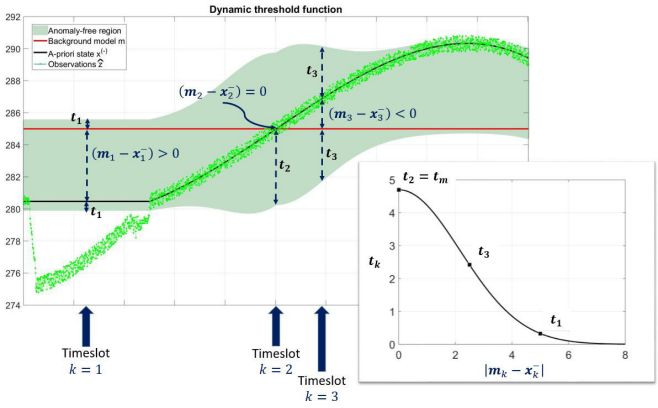


Fig. 1. Dynamic threshold function, example. Graphical representation of (9) and (10). The green dots correspond to the observations $\hat{\boldsymbol{z}}_k$, the black curve is the *a priori* estimate $\boldsymbol{x}_k^-$, and the red curve represents the background model $\boldsymbol{m}_k$. The green area corresponds to the anomaly-free region. A detailed description is provided in the text. (Bottom left) Example of (10), using the values in Table III for the IR039 channel.

A nominal scenario is detected if the following condition on the observation $\hat{z}_k$ is verified:

$$\hat{z}_k \in \left[\min\left(\boldsymbol{m}_k, \boldsymbol{x}_k^-\right) - \boldsymbol{t}_k, \max\left(\boldsymbol{m}_k, \boldsymbol{x}_k^-\right) + \boldsymbol{t}_k\right]. \tag{9}$$

The observation must reside within the region delimited by the background model and the Kalman *a priori* estimate plus and minus a dynamic threshold value, which once again depends on the difference between the background model and the *a priori* estimate. The single element $\{t_i\}_k$ of the threshold vector $\boldsymbol{t}_k$ vector is given by

$$\{t_i\}_k = t_{mi} e^{-\frac{\left(\{x_i\}_k^- - \{m_i\}_k\right)^2}{2\sigma_i^t}} \tag{10}$$

where t_{mi} is the maximum threshold assigned to the channel i and σ_i^t represents the confidence interval in which the *a priori* state can still be representative of a nominal situation. Reasonable approximation is to consider $\sigma_i^t \cong t_{mi}$, as in most of the cases they physically represent the same quantity, and cannot exceed the latter by definition.

Fig. 1 provides a graphical representation of (9) and (10) to clarify and better explain the rationale behind such formulation. As evident in Fig. 1, (9) defines an anomaly-free region where a given observation $\hat{\boldsymbol{z}}_k$ can be considered nominal (i.e., not an anomaly.) Such region varies dynamically with time and exclusively depends on the *a priori* Kalman estimate and the background model, at a given timeslot. For instance, the timeslot $k = 1$ in Fig. 1 refers to a case where the *a priori* estimate is below the actual pixel model (fixed) at the same timeslot. In this particular case, as the difference between the two is relatively large, the dynamic threshold in (10) results being small (value $\boldsymbol{t}_1$ in Fig. 1). All new observations falling within the anomaly-free region (green area) would therefore be classified as nominal and exploited by the filter to update its status. The same considerations hold for the timeslot $k = 3$, where the only differences reside in: 1) the *a priori* state estimate is above the relevant background model value and 2) the difference between $\boldsymbol{m}_k$ and $\boldsymbol{x}_k^-$ is smaller and the $\boldsymbol{t}_3$ value is indeed larger. The timeslot $k = 2$ presents a slightly

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different scenario, as the *a priori* Kalman estimate corresponds to the background model (ideal situation); hence, the value of t_2 is the maximum possible ($t_2 = t_m$) and defines fully by itself the anomaly-free region.

The Gaussian function in (10) is selected to allow smooth transitions among time steps, and to guarantee a continuous behavior of the dynamic threshold function (as evident in Fig. 1) Moreover, a Gaussian expression allows the KAM algorithm to be easily tailored for any application, as the only necessary predefined (input) vector is t_m (e.g., using a triangular function would need a retuning of the whole threshold function formulation depending on the application, other than provoking possible discontinuities in the anomaly-free region trend.) Also, it is to be noted how the Kalman status would change only when an observation is accepted as anomaly-free: in the example in Fig. 1, some measurement examples (close to the timeslot $k = 1$) are identified as anomalies. Due to this, the filter does not change its status and continue propagating the relevant information maintained from the last useful nominal timeslot (however, adding *a posteriori* noise to the error covariance, not visible in Fig. 1). Further details are provided in the following paragraphs and in the chapter devoted to the application. The definition of the dynamic threshold (and associated anomaly-free region) for anomaly detection is one of the elements of novelty of the proposed technique.

D. Kalman Gain Computation and A Posteriori State

The Kalman gain K_k at time step k is then computed depending on the outcome of the anomaly detection. Practically speaking, the information provided by the actual observation is taken into account only if conditions were nominal. Contrary, the associated *a priori* information is exclusively propagated to compute *a posteriori* estimates, such as the boundary conditions for the next timeslot. The Kalman gain matrix K_k ($n \times n$) is defined as follows:

$$\begin{cases} K_k = P_k^- \left(P_k^- + R_k\right)^{-1}, & \text{if nominal state} \\ K_k = \mathbf{0}, & \text{if anomaly state} \end{cases} \tag{11}$$

where $\mathbf{0}$ stands for the zero matrix ($n \times n$). The *a posteriori* state vector and covariance matrix are obtained by combining all the considered quantities. They provide the initial conditions for the timeslot $k + 1$ by properly weighing the available information depending on the associated noise. The *a posteriori* state vector x_k^+ ($n \times 1$) and the *a posteriori* error covariance matrix P_k^+ ($n \times n$) are computed by means of [9]

$$\begin{cases} x_k^+ = x_k^- + K_k(\hat{z}_k - x_k^-) \\ P_k^+ = (I - K_k)\, P_k^- \,(I - K_k)^T + K_k R_k K_k^{\mathrm{T}} \end{cases} \tag{12}$$

where I is the identity matrix. The P_k^+ expression is known as the Joseph form of the covariance update equation and can be applied with any value of K_k. Such formulation comprises other simplified forms and provides the best numerical computation properties as well as helps with reducing numerical errors [11].

III. Geostationary Multispectral Data Processing

The proposed KAM algorithm is based on the temporal Kalman filter, used as a detection system, with the main goal of developing a nearly real-time operational application. The latter can be provided, for instance, by fixed-pointing ground-based multispectral video cameras or by geostationary remote-imaging sensors. The latter observation systems can guarantee a proper and recurrent time sampling, and a fixed spatial characterization of the observed scenario.

In this work, we have decided to use the geostationary MSG satellite with on board the so-called Spinning Enhanced Visible and Infrared Imager (SEVIRI), which has 12 spectral channels [16]. Earth imaging is obtained by a bidimensional Earth scan, combining the satellite spin and the scan mirror rotation. In this study, the rectified MSG SEVIRI image data are used. The SEVIRI Level 1.5 product is the result of a sequence of radiometric processing performed on the raw SEVIRI data such as linearization, conversion into radiances, calibration, and linear scaling [28], [29]. SEVIRI observes the Earth–atmosphere system with a spatial sampling distance of 3 km at the equator. The actual instantaneous field of view of the channels is about 4.8 km (11 channels) at nadir. The spatial resolution increases as one moves further from the subsatellite point. In particular for the chosen area of study in Western Africa, the ground resolution of the SEVIRI's 12-channel imager range from 3.1 to 4 km in both N–S and E–W directions [16]. SEVIRI takes images of the Earth at regular intervals, during a 15-min image repeat cycle (involving a 12-min 30-s Earth imaging phase and an up to 2-min 30-s calibration and retrace phase); thus, resulting in a 15-min temporal resolution (corresponding to time interval in between two consecutive timeslots of the KAM).

Data are acquired in near real time under the high rate information transmission (HRIT) format which is the standard adopted by EUMETSAT to encode Level 1.5 SEVIRI image data [25]. Such HRIT files are ordered via the EUMETSAT Data Centre [32], in a lossless wavelet compressed file format. The wavelet decompression is then performed using the "Public Wavelet Transform Decompression software" provided by license by EUMETSAT. Concerning the navigation of image data, the conversion of image coordinates into geographical ones is performed through the combination of a linear scaling function and a nonlinear projection function [24]. Such procedure is solved through the creation of a "Navigation" GeoTIFF file starting from the border pixels of the MSG SEVIRI Native grid with the use of coefficients for the conversion in GEOS projection [26]. In this way, the resulting latitude and longitude of a pixel centroid referring to image coordinates of the original MSG grid (full-disk) are directly obtained.

Only five out of the 12 spectral SEVIRI channels have been adopted, which are the VIS 0.6μm and the IRs at 3.9, 10.2, 12.0, and 13.4 μm. All spectral measurements have been calibrated. For the VIS channel, the radiance has been converted in [mW·m^{-2}·sr^{-1} (cm^{-1})$^{-1}$], while all IR channels have been treated in K. The radiance R_{λ_i} expressed in [mW·m^{-2}·sr^{-1}(cm^{-1})$^{-1}$] can be converted in radiance R_{m_i} in [W·m^{-2}·sr^{-1} · μm^{-1}] by means of $R_{m_i} = \left(10/\lambda_i^2\right) R_{\lambda_i}$.

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Furthermore, to mask out the twilight effect in VIS channels a dedicated masking procedure have been implemented by setting a limit into the solar zenith angle θ_s. In particular, to filter out reflectance anomalies over the day–night transition area of a particular scene, only pixels having Sun zenith angle θ_s lower than 80° are taken as valid by the algorithm [26]. The solar zenith angle θ_s is function of time and location can be estimated using the following equation:

$$\cos\theta_s = \sin\phi\sin\delta + \cos\phi\cos\delta\cosh_a \tag{13}$$

in which δ is the declination of the Sun, ϕ is the latitude, and h_a is the hour angle.

The period of analysis goes from December 2015 to the end of February 2016. The data set is uniformly sampled every 15 min. The algorithm has been applied on the countries of the Economic Community of West African States (ECOWAS) countries plus Mauritania and Chad. The area of interest has an top-left corner of 1.355° N, 24.725° E and bottom-right corner of 27.027° N, 20.942° W, and MSG image segments 5, 6 of MSG full-disk product are processed and enough to cover such area. To characterize SEVIRI data only over land, a water mask in MSG full disk grid has been derived for the entire area of interest using the cloud mask (CLM) product provided by EUMETSAT in GRIB2 format [23]. In particular, the resulting water mask binary GeoTIFF file is derived by extracting all pixels classified as "Clear sky over water" from all CLM products available over the same grid in the entire period of analysis. It is noted that the observed scene is kept under the original geostationary grid and covers 1530 × 880 pixels for each considered timeslot that is 1 346 400 pixels. Water pixels constitute the 18.12% (243 967 pixels) of the entire scene and are discarded for our purposes. The total number of pixels over land is indicated with N_{pixel}, equal to 1 102 433 pixels. The number of timeslots per day for the entire period is indicated with D_{day}, where $D_{\text{day}} = 96$. Only five timeslots have been discarded after a proper data quality assessment due to Sun collinearity with the MSG satellite (February 27, 2016 12:00, February 28, 2016 12:00, February 28, 2016 12:15, February 29, 2016 12:00, February 29, 2016 12:15.)

IV. Application of Kalman-Based Clear-Air Masking

The KAM approach is here fine-tuned and applied to identify scenarios of clear-air conditions, that is defining the anomaly as anything that cannot be categorized as clear sky.

Fig. 2 shows the overall filter architecture going from the available data set (described in Section III), the actual measurement, and the relative background model are injected as inputs to the filtering algorithm. A water mask is applied to filter out pixels covered by water (not considered in this work). The clear-air KAM workflow follows the step defined in Section II and provides the information whether the observed pixel is in pure clear-air condition or an anomaly is detected. This application represents the initial step in many space-borne remote-sensing applications. A future effort may be devoted in performing a classification of the detected anomaly making use of the Kalman states complemented by the information provided by the background model. Applications can be manifold, as fire detection, volcanic ash detection or any other phenomenon that presents a distinct spectral behavior at the wavelengths of interest.

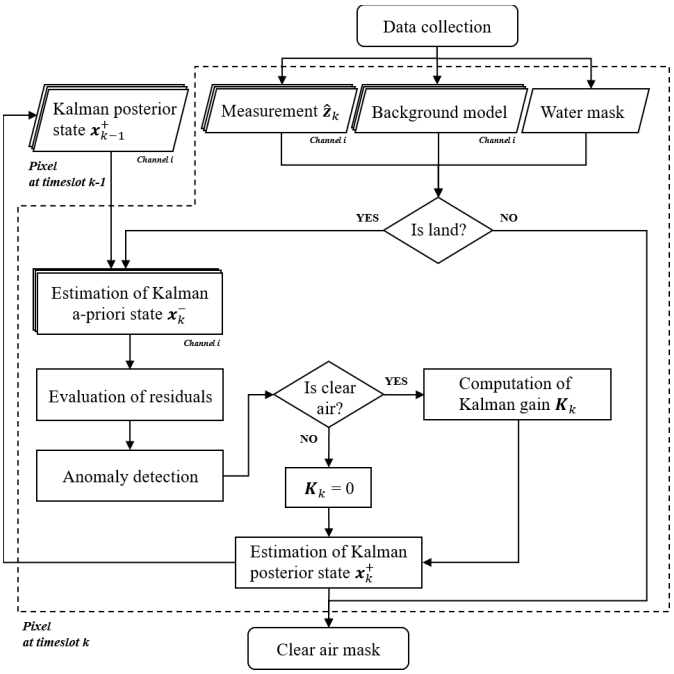


Fig. 2. Schema of the multispectral Kalman-based clear-air masking (KAM) filter. Each module corresponds to a software component developed for nearly real-time applications of the algorithm for clear-air masking.

The next sections describe the tailoring of the KAM filter for the clear-air masking application.

A. Observations and Noise Covariances

As mentioned, the actual observation vector $\underline{\hat{z}}_k$ $(n \times 1)$ describes a scenario characterized by several representations (multispectral observations). In particular, it is formed by the following MSG-SEVIRI channels (as described in Section III): the VIS 0.6μm and the IRs at 3.9, 10.8, 12.0, and 13.4 μm; thus, $\hat{z}_k = \left(\hat{z}_{0.6\ \mu\text{m}}, \hat{z}_{3.9\ \mu\text{m}}, \hat{z}_{10.8\ \mu\text{m}}, \hat{z}_{12.0\ \mu\text{m}}, \hat{z}_{13.4\ \mu\text{m}}\right)_k^T$. VIS channel measurements are expressed in terms of reflectance in $[\text{mW}\cdot\text{m}^{-2}\cdot\text{sr}^{-1}\cdot(\text{cm}^{-1})^{-1}]$, while all IR channels are converted in K. The index k indicates the considered timeslot.

The covariances among noise pairs at the timeslot k form the observation noise covariance matrix R_k $(n \times n)$. Practically speaking, the latter matrix describes the correlation among errors associated with the various SEVIRI channels [see (3)]. It is common practice in data assimilation to treat observation errors as uncorrelated, that is interchannel error correlations can be assumed null ($\boldsymbol{R}_k$ diagonal matrix). This is often an instrument design specification and, in the case of SEVIRI, studies have been performed to better characterize this aspect [1]. On the other hand, it is clear that the elements on the principal diagonal correspond to the error variance associated with each of the SEVIRI channels [17], [18].

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TABLE II

SEVIRI NOISE BUDGETS AS MEASURED AT THE BEGINNING OF LIFE, EXPECTED AT THE END OF LIFE, AND THE SPECIFICS [16]–[18]

Channel	VIS06 0.6 μm [W m^{-2} sr^{-1} μm^{-1}]	IR039 3.9 μm [K]	IR108 10.8 μm [K]	IR120 12.0 μm [K]	IR134 13.4 μm [K]
Measured noise at the beginning of life	0.39	0.24	0.13	0.21	0.29
Expected noise at the end of life	0.59	0.36	0.20	0.32	0.44
Noise (specification)	**0.53**	**0.35**	**0.25**	**0.37**	**1.80**

The variance can be computed as the square of absolute radiometric accuracy and can be retrieved from the instrument specifications, retrieved from [16]–[18], and reported in Table II. The three rows correspond to the SEVIRI measurement uncertainty predicted at the beginning and at end of the mission life; as well as the imposed specifications at design level. The specifications were considered as baseline in the KAM application (worst case scenario.)

B. Background Model Characterization

One of the essential step of the KAM algorithm is the background model characterization for each of the employed MSG SEVIRI channel. A good channel-background model of a certain pixel must be purely representative of a clear-air situation with as less as possible associated noise, which is necessary for an accurate anomaly detection. In this work, we adopted static background models only and future studies are foreseen to refine the approach and consider a dynamic update of the model (through a combination of static models and a time-based moving average system).

Cloud-free scenes are identified by using the CLM product [23] projected in MSG full disk grid over the period going from December 2015 to the end of March 2016. For each of the considered timeslots (96 timeslots in a day) and relevant channel, an image stack was considered after filtering out cloudy scenarios. Average and standard deviation (1-sigma) are then computed for each stack at pixel level to obtain the background model quantities, $\{m_i\}_k$ and $\{\sigma_i^m\}_k$, respectively; one for each of the five SEVIRI channels and for each of the 96 timeslots. It is to be noted that the computed background models are completely general and mediated over the period, and do not represent the specific daily trend to be expected over a certain period (i.e., they do not include any short-term effect.) The background model vector of the considered scenario $\boldsymbol{m}_k$ $(n \times 1)$ is formed by the mean values of the stacks (one per channel) at a given timeslot k (i.e., 15-min sampling.) While the associated noise vector $\boldsymbol{\sigma}_k^{\boldsymbol{m}}$ $(n \times 1)$ is formed by the standard deviations of the stacks (one per channel) at a given timeslot k and quantifies the expected variations of the channel radiances under clear sky conditions on the long term.

TABLE III

THRESHOLD VALUES FOR ANOMALY DETECTION WHEN USING MSG-SEVIRI

Channel	VIS06 0.6 μm [W m^{-2} sr^{-1} μm^{-1}]	IR039 3.9 μm [K]	IR108 10.8 μm [K]	IR120 12.0 μm [K]	IR134 13.4 μm [K]
t_{mi}	2.3	4.7	4.5	3.9	3.5

C. Kalman-Based Filter Tailoring

Referring to the KAM filter depicted in Section II, the clear-air anomaly detection is the only module that needs to get tailored for the adopted instrument. In particular, we would need to define the proper elements of the $\boldsymbol{t}_m$ vector $(n \times 1)$ to be used in (10). In general, the smaller the t_{mi} values, the more stringent the anomaly detection algorithm. Small t_{mi} values would therefore make the Kalman algorithm very selective in accepting a clear-air condition, which would be anyway the preferred option for a filter propagation perspective. If the t_{mi} values are too large, the filter would wrongly associate an anomaly scenario as clear-air, resulting in a propagation of the wrong information to the *a posteriori* status and therefore to the following timeslot.

The applied solution was to compute the t_{mi} values making use of the information provided by the background model associated noise, hence becoming the only element needed to run the algorithm. In particular, the standard deviations $\{\sigma_i^m\}_k$ associated with the background models were exploited to compute the $\boldsymbol{t}_m$ vector $(n \times 1)$. For each SEVIRI channel $i = 1, \ldots, n$ $(n = 8)$, the t_{mi} value was computed as follows:

$$t_{mi} = \frac{1}{(D_{\text{day}} + N_{\text{pixel}})} \sum_{N_{\text{pixel}}} \left[\sum_{k=1}^{D_{\text{day}}} \{\sigma_i^m\}_k \right] \tag{14}$$

where D is the number of timeslots per day ($D_{\text{day}} = 96$ in this application) and N_{pixel} is the total number of pixels over land composing the observed scene. The $\boldsymbol{t}_m$ vector is therefore independent of time. Table III recaps the parameters of interest used in (10).

It is worth noting that such thresholds are applied indistinctively over the whole scene, and only depends on the following:

1) the considered wavelength and its sensitivity to the considered anomaly, as well as the *a priori* estimate deviation in clear sky conditions—meaning that the values represent the expected maximum interval within which we can still consider the observed scenario being in clear-air conditions;
2) the adopted instrument, and therefore its sensitivity to the scene at a given wavelength, other than the considered spatial resolution.

The most important asset of the proposed KAM approach is the possibility to define a set of thresholds to be independently applied. The pixel-specific information is completely absorbed by the *a priori* Kalman estimate by exploiting the background model (e.g., no need of having different thresholds

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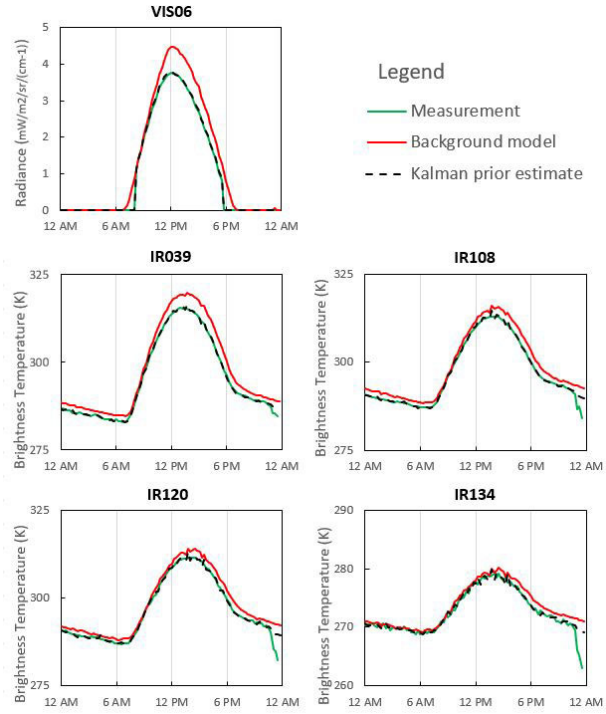


Fig. 3. Example of clear-air day (December 9, 2015) for all considered channels of the MSG-SEVIRI instrument. The green curves represent the actual observation vector $\hat{z}_k$, the black dotted curve is the *a priori* estimate $\boldsymbol{x}_k^-$, and the red curve represents the background model $\boldsymbol{m}_k$. A detailed description is provided in the text.

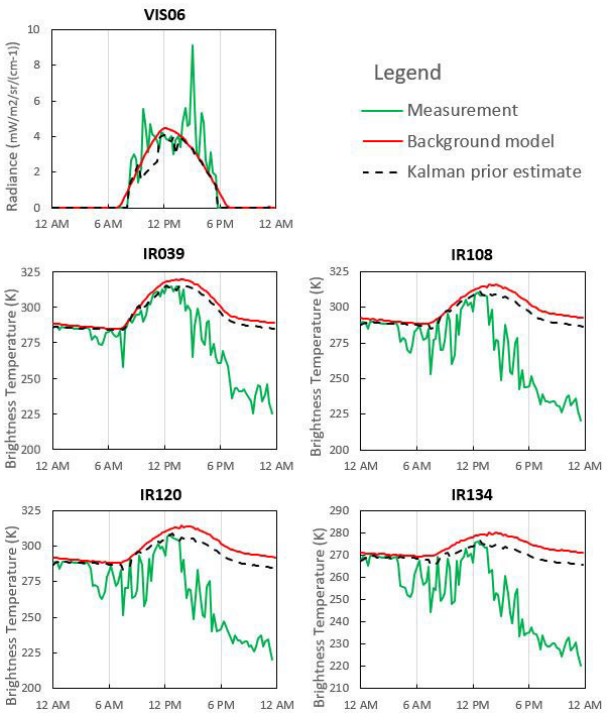


Fig. 4. Example of "anomaly" day (December 10, 2015) for all considered channels of the MSG-SEVIRI instrument. The green curves represent the actual observation vector $\hat{z}_k$, the black-dotted curve is the *a priori* estimate $\boldsymbol{x}_k^-$, and the red curve represents the background model $\boldsymbol{m}_k$. A detailed description is provided in the text.

to be applied in daily or night conditions as in most of the contextual-spatial detection algorithms).

D. Clear-Air KAM Application

The algorithm described in Section II has been applied to the available data set presented in Section III over land pixels. The entire computation relies on a chain developed in Python, which has been developed by Progressive Systems, Rome, Italy, in the context of the PROBA-V MEP TPS research project funded by ESA/ESRIN Frascati RM, Italy [34]. The processing here presented has been performed in the computer resources offered within the Research and Service Support (RSS) CloudToolbox service [33] managed by the ESA's RSS [27].

We report here some examples of temporal trends, with the description of the time evolution of the associated key quantities governing the KAM algorithm. Fig. 3 shows the KAM filter behavior during a clear-air day (December 9, 2015) for a sample pixel in the Senegal area and for all adopted MSG-SEVIRI channels. In such example, the measurements (observations $\hat{z}_k$, red curves) clearly describe clear-air conditions over the whole day. Hence, the *a priori* estimates ($\boldsymbol{x}_k^-$, black-dotted lines) tend to follow the measurements by exploiting the state transition matrix computed from the background model values ($\boldsymbol{m}_k$.) Small signal variations of the *a priori* estimates can be observed and such differences are sequentially corrected when updating the *a posteriori* estimates (not shown) with the actual measurements so that, at the next state, the most reliable information is used to propagate the filter status. This is the best possible situation as the KAM is relying on useful measurements to continuously update its status with the minimum possible associated noise (basically corresponding to the instrument noise exclusively.) We can also observe how, around 10:00 P.M., the presence of a light cloud decreases the brightness temperatures for some of the considered channels. The KAM filter is able to detect an anomaly behavior thanks to the information provided by the channels at 12 and 13 μm (most affected), confirming the importance of having a multispectral anomaly detection module. The *a priori* estimate maintains the information provided by the *a posteriori* one and does not follow the non-clear-air observed value; here the filter does not exploit the actual measurements but rather propagates the estimates available from the previous slot adding a higher associated noise (making use of the process noise covariance matrix.)

Fig. 4 better highlights the mentioned behavior, where the daily trend of the same pixel is analyzed for the following day (December 10, 2015). The first half of the day is characterized by a discontinuous presence of light cloudy conditions, several anomalies are identified, and the *a priori* estimates always provide a consistent clear-air-like trend as they should (by correctly ignoring the measurements that do not fall within the multispectral anomaly-free region.) When a clear-air observation is available (few timeslots during the first part of the day), clear-air scenarios are detected and used by the filter to update the *a posteriori* estimate accordingly. In the afternoon, all considered channels reflect a

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persistent and heavier cloud coverage, making the information coming from the actual observation vector useless for what concerns the filter propagation. Nevertheless, the KAM filter continues providing an equivalent clear-air estimate basing the calculation on the last available measurements accepted by the filter and by sequentially adding the associated noise to the *a posteriori* estimate (i.e., taking into account the noise associated with the state transition matrix.)

As already mentioned, the KAM detection module is applied by looking at all available channels simultaneously, making the filter able to sound the atmosphere at different wavelengths and to account for all possible anomaly scenarios. This allows excluding several false alarms or missed detections, since one channel may be less sensitive than others to an anomaly situation (or vice versa). The continuous availability of the *a priori* estimates $\boldsymbol{x}_k^-$ of the observed scene (computed exploiting the previous Kalman filter states and the pixel models) allows having an equivalent clear-air radiance for all channels, expected to be measured in the absence of anomaly (could coverage.) Exploiting this information would be very beneficial for several retrieval methods of secondary products associated with the observed scenario, being therefore able to isolate the contribution of the nominal radiance. Future works will specifically address the matter.

V. Validation

In this section, we present an intercomparison of the presented algorithm with respect to the EUMETSAT cloud-masking product. The latter has been extensively validated and operationally used in several real-time applications over the last decade; it therefore provides a solid comparison term for the validation [23]. Moreover, the selected comparison allows validating the clear-air KAM filter using the same input data (the same MSG SEVIRI geostationary grid is considered) to minimize any resampling of the product which may cause a loss of useful information. There is a fundamental difference between the two data sets: while the KAM approach is entirely temporal-based, the EUMETSAT cloud-masking algorithm only exploits spatially-based techniques by setting several thresholds and referring them to the "mean" value of a moving spatial average window centered at the pixel of interest [23].

The period of analysis goes from December 2015 to the end of February 2016. The data set is uniformly sampled every 15 min. The observed scene is kept under the original geostationary grid and covers 1530 × 880 pixels for each considered timeslot that is 1 346 400 pixels. Water pixels constitute the 18.12% of the entire scene and are discarded for our purposes. The 96 timeslots per day are available for the entire period. For the validation, we define four classes: 1) blue and cyan dots refer to situations of matching in clear-air and anomaly conditions, respectively; 2) yellow dots indicate missing detections of clear-air by the KAM filter with respect to the EUMETSAT CLM; and 3) finally, red dots show false alarms of clear-air.

In some conditions, the KAM and EUMETSAT algorithms may lead to different results because the Kalman-based technique only exploits the image temporal changes and is

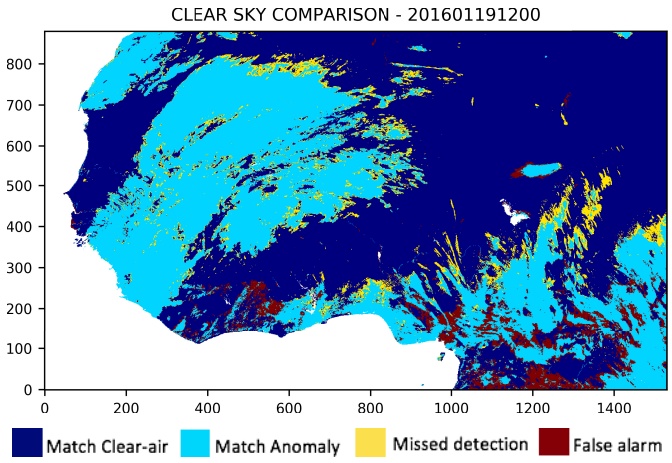


Fig. 5. Intercomparison with respect to EUMETSAT CLM product for a single timeslot (January 19, 2016 12:00 h). Blue and cyan points correspond to pixels for which the two algorithms provide the same result (clear-air and anomaly, respectively.) Yellow points represent missed detections of the KAM algorithm with respect to the EUMETSAT masking product, while brown pixels indicate false alarms.

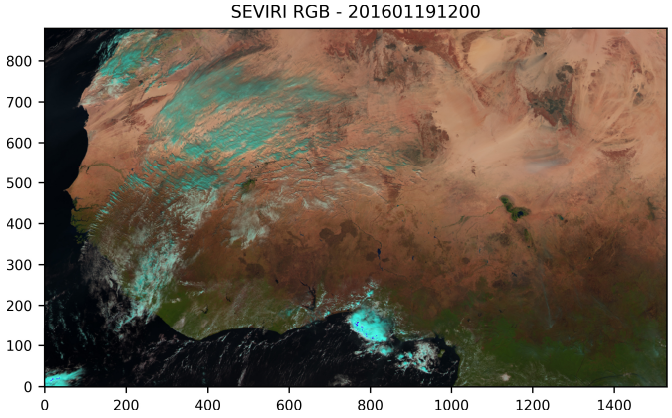


Fig. 6. Composite RGB image from the MSG-SEVIRI instrument for the considered timeslot in Fig. 5 (January 19, 2016 12:00 h).

therefore independent of common biases potentially affecting the spatial pixel grid used by EUMETSAT. Moreover, while we aim at detecting pure situations of clear-air conditions (clear-air masking), the EUMETSAT product is intended to provide a cloud-masking product. Practically speaking, it is much more important for our validation purposes to minimize the missed detections of clear-air conditions—cases when the KAM algorithm indicates clear-air whereas the EUMETSAT one indicates the cloud presence. False alarms are less critical: as the EUMETSAT product may flag a certain pixel as cloud free, but the Kalman-based detection system could classify that particular condition as a scenario not purely clear-air representative. It is also important to underline that both algorithms do not provide any ground truth and differences may be related to stricter or larger thresholds other than limits linked to such approaches.

Fig. 5 shows an example of comparison for the whole area under analysis and for a single timeslot (December 13, 2015 12:00 h). For visual interpretation, Fig. 6 provides the

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TABLE IV
INTERCOMPARISON WITH RESPECT TO EUMETSAT CLM PRODUCT

CLEAR-AIR PIXELS (N. SAMPLES = 6.8407E+09)		CLOUDY/ANOMALY PIXELS (N. SAMPLES = 2.7843E+09)	
MATCH	MISSED DETECT.	MATCH	FALSE ALARMS
95.7241 %	4.2759 %	74.9919 %	25.0081 %

ALL PIXELS (N. SAMPLES = 9.6250E+09)		
MATCH	MISSED DETECTIONS	FALSE ALARMS
89.7267 %	3.0390 %	7.2343 %

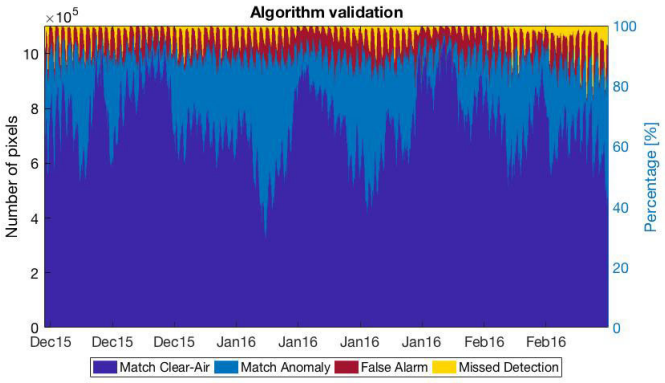


Fig. 7. Intercomparison (temporal trend) between the KAM algorithm and the EUMETSAT CLM product over the whole analyzed period for all pixels composing the scene. Blue and cyan points correspond to pixels for which the two algorithms provide the same result (clear-air and anomaly, respectively.) Yellow points represent missed detections of the KAM algorithm with respect to the EUMETSAT masking product, while brown pixels indicate false alarms.

corresponding composite RGB image derived from the VIS and near-IR SEVIRI channels. A fairly good agreement is globally reached and yellow/red areas mainly correspond to borderline situations. Some mismatches, especially false alarms, are geographically grouped in the lower part of Fig. 5/Fig. 6. On top of the above mentioned considerations, this particular behavior might be related to a relatively poor characterization of the background pixel models. Exploiting longer time series would certainly overcome the issue and improve the final result.

Table IV provides the percentages of matching, missed detection and false alarm over the entire test period. The percentages take into account the whole scene, as well as differentiating the calculations on clear-air and anomaly scenarios. Table IV also reports the total number of pixels considered for the validation. The missed detections are very limited to about 3% over the entire period and mainly due to mischaracterizations of the background models. False alarms are more often registered, however, less critical for the reasons explained above.

Fig. 7 shows the same comparison in terms of temporal trend of both the percentages and number of pixels for each single timeslot. The same color legend is adopted: dark blue refers to clear-air situations, light blue to anomaly situations, yellow to missed detections, and brown to false alarms. It can be seen how the total matching (dark blue plus light blue) is mainly constant over the whole period of analysis with a percentage of absolute matching placed around 90%.

VI. CONCLUSION

An anomaly detection and estimation technique has been presented, using a modified version of Kalman filter on a pixel basis in the temporal domain. Advantages of using a time-based KAM approach have been discussed, blazing a trail to many applications in detecting and monitoring natural and environmental phenomena around the world. After presenting the KAM system in a general framework, an application of the latter has been shown to prove its performance in detecting clear-air scenarios.

The clear-air KAM algorithm has been applied to multispectral observations available from geostationary VIS-IR radiometric measurements coming from SEVIRI instrument on board of the Meteosat satellite. The area of interests covers the whole West Africa, for a test period of three months from December 2015 till the end of February 2016. This results in a large amount of processed pixels having 96 timeslots per day. A validation of the clear-air algorithm has been presented by intercomparing the detection results with the well-known EUMETSAT CLM product. This has ensured to have the same input data processed by a completely different algorithm approach. The validation has shown constant percentages of matching around 90% over the entire period of analysis.

The add-on value of the Kalman approach is to be able to provide *a priori* estimates of the scenario, exploiting the previous Kalman filter states and making use of the pixel models. This allows having an equivalent clear-air radiance which is expected to be measured in the absence of anomaly (for instance, in the absence of cloud coverage). In such a way, the KAM algorithm can be globally applied, since the estimates are computed from the pixel models and the thresholds can be set independently of the area of interest.

Future work will address the use of the available *a priori* states of the Kalman filter to develop atmospheric correction methods through the estimation of the atmospheric optical thickness. Moreover, the efforts will be devoted to fine tuning of the KAM algorithm to add robustness (e.g., including channel weights in the anomaly detection) as well as to extending the application over other regions for further validation.

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